London, UK (on a Global Talent Visa)
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Egor Burkov

Profile

- Computer vision researcher with a PhD: pose estimation, interactive avatars, face synthesis, diffusion for image generation, neural 3D reconstruction, 3D vision.
- Looking to dive into cutting-edge vision/3D (e.g. video generation, world models) as a researcher; also interested in growing towards roles involving more communication/planning/ownership.

Work

2025– **Avaturn (a.k.a. Yolk)**, *Research Scientist*
Built AVTR-1 – real-time (streaming) avatars with active listening, 3rd in NVIDIA StreamEval benchmark.
Bonus project: LLM-judge-based evals for an agentic "AI sales copilot".

2023 **Meta Reality Labs – Zürich**, *Research Scientist Intern*
Retrained large models for image generation (Emu) to run faster (3 diffusion steps instead of 6-8).

2021–23 **Skoltech**, *Research Sciewntist*
Created Multi-NeuS, a neural implicit function for 3D human head reconstruction from one photo.

2018–21 **Samsung AI Center – Moscow**, *Telepresence lab, Research Scientist*
Head animation via latent pose project (CVPR 2020): original idea, most experiments, led the team.
3D human pose estimation from single-view video (part of a bigger AR system): developed personally.
3D human pose estimation from multiple views: responsible for data and minor stuff.
Extra: running neural nets on Android, viz tools, data, open-sourcing...

2015–17 **VisionLabs**, *Research Engineer*
OpenCV interface for Torch (Lua), open source – developed personally.
Re-trained face recognition neural nets to reduce runtime and parameter count.
Extra: facial keypoints detection on Android, 3DMM fitting, prototype demos...

Education

2018–24 **PhD in Computer Science**, *Skoltech, supervised by Victor Lempitsky*
Thesis: *Learning from Data for Human Modeling and Tracking*.
Taught 8 and authored 3 courses. Designed auto-graded Deep Learning home assignments from scratch.
Built+managed research group's DIY GPU cluster. Presented own entrepreneurship project at SLUSH.

2016–18 **MSc in Computer Science**, *Skoltech*, with distinction
Thesis: *Deep Neural Networks with Box Convolutions* (NeurIPS 2018, $\star$ 500+).

2012–16 **BSc in Computer Science**, *HSE University*
Thesis: *ConvNet-based Human Segmentation Using Background Subtraction Map.*

Selected Publications

- **Neural Head Reenactment with Latent Pose Descriptors**. CVPR 2020. **E. Burkov et al.**
- **Multi-NeuS: 3D Head Portraits from Single Image with Neural Implicit Functions**. IEEE Access 11, 2023. **E. Burkov et al.**
- **Learnable Triangulation of Human Pose**. ICCV 2019. **K.Iskakov et al.**
- **Deep Neural Networks with Box Convolutions**. NeurIPS 2018. **E. Burkov, V. Lempitsky**
- **Textured Neural Avatars**. CVPR 2019. **A. Shysheya et al.**

Open-Source Software

- **Box convolution layer for PyTorch**. ⋆ $0.5k$
- **Head reenactment with latent pose descriptors**. ⋆ $0.2k$
- **Learnable human pose triangulation**. ⋆ 1.1k
- **OpenCV interface for Torch (Lua)**. ⋆ $0.2k$
- **AVTR-1 – real-time interactive avatars with active listening**. ⋆ $0.5k$

Everything Else

2024– **Small business owner (offline)**

A skateboarding school: 2+ years, 1600+ classes, 9 employees. Went from doing everything myself (coaching, hiring, building, media/ads, sales, events, customer service) to established processes with complete delegation.

November 2026